Adı ve Soyadı:
No:

2025-2026 Öğretim Yılı Güz Dönemi
MÜHENDİSLİK FAKÜLTESİ GENEL MATEMATİK I BÜTÜNLEME SINAVI
(Süre 80 dk.)

1. $\int \dfrac{\sin^3 x}{1-\cos x}dx = ?$ (20 p)

$$I = \int \frac{\sin^2 x \cdot \sin x}{1-\cos x}dx = \int \frac{(1-\cos^2 x)\cdot \sin x}{1-\cos x}dx = \int \frac{(1-\cos x)(1+\cos x)\cdot \sin x}{1-\cos x}dx$$

$$= \int (1+\cos x)\cdot \sin x \cdot dx \quad \begin{pmatrix} 1+\cos x = u \\ -\sin x dx = du \end{pmatrix} \quad 10$$

$$= -\int u \cdot du = -\frac{u^2}{2} + C = -\frac{1}{2}(1+\cos x)^2 + C \quad 10$$

2. $\int e^x \ln(1+e^x)dx = ?$ (20 p)

$1+e^x = t \Rightarrow e^x dx = dt \Rightarrow I = \int \ln t \cdot dt$ 5

$\ln t = u \Rightarrow \frac{1}{t}\cdot dt = du$, $dt = dv \Rightarrow v = t$ 5

$$I = u.v - \int v.du = t.\ln t - \int t.\frac{1}{t}\cdot dt \quad 5$$

$$= t.\ln t - t + C$$

$$= (1+e^x).(\ln(1+e^x) - 1) + C \quad 5$$

3. $y = xe^x$ fonksiyonunu minimum yapan x değerini bularak, bu noktadan geçen teğet denklemini yazınız. (20 P)

$\frac{dy}{dx} = e^x + xe^x = e^x.(1+x) = 0 \Rightarrow x = -1$ K.N.

(Y.min.) 8

x		-1	
y'	−	0	+
y	↘	-1/e	↗
		min	

$y - y_0 = m.(x - x_0)$

$x_0 = -1 \Rightarrow y_0 = -1.e^{-1} = -\frac{1}{e}$

$m = f'(x_0) = e^x.(1+x)\big|_{x=-1} = 0$ } 6

$y - (-\frac{1}{e}) = 0.(x-(-1))$

$y = -\frac{1}{e}$ 6

4. $\lim\limits_{x\to\infty}\left(1-\sin\left(\frac{1}{2x}\right)\right)^{6x}=?$ (15 p)

$y=\left(1-\sin\frac{1}{2x}\right)^{6x} \Rightarrow \ln y = 6x\cdot\ln\left(1-\sin\frac{1}{2x}\right)$ 2/

$\lim\limits_{x\to\infty}\ln y = \lim\limits_{x\to\infty}\dfrac{6\cdot\ln\left(1-\sin\frac{1}{2x}\right)}{\frac{1}{x}}$ $\left(\frac{0}{0}\right)$ $\left(x\to\infty \Rightarrow \frac{1}{x}\to 0\right)$

$= \lim\limits_{x\to\infty}\dfrac{6\left(+\frac{1}{2}\cdot\frac{1}{x^2}\cdot\cos\frac{1}{2x}\right)}{1-\sin\frac{1}{2x}}$ / $-\frac{1}{x^2}$

$=-3$ 10/

$\lim\limits_{x\to\infty}\ln y = -3$ olup 3/

$\lim\limits_{x\to\infty} y = e^{-3}$ dur.

5. $y=\dfrac{e^x}{x}$ eğrisini gerekli incelemeleri yaparak çiziniz. (25 p)

- $T=R-\{0\}$ 3/
- $\lim\limits_{x\to+\infty}\dfrac{e^x}{x}=\lim\limits_{x\to+\infty}e^x=\infty$

$\lim\limits_{x\to-\infty}\dfrac{e^x}{x}=\dfrac{e^{-\infty}}{-\infty}=\dfrac{0}{-\infty}=0$ 5/

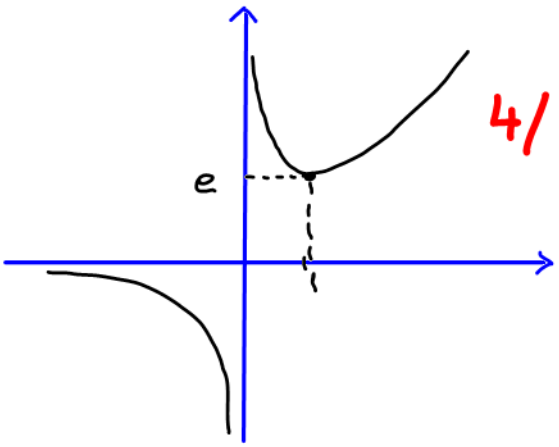
($y=0$ Yatay Asmp.)

$\lim\limits_{x\to0^+}\dfrac{e^x}{x}=\dfrac{1}{\neq0^+}=+\infty$

$\lim\limits_{x\to0^-}f(x)=\dfrac{1}{\neq0^-}=-\infty$ 5/ ($x=0$ Düşey Asmp.)

- $y'=\dfrac{xe^x-e^x}{x^2}=\dfrac{e^x\cdot(x-1)}{x^2}=0$ 4/

($x=1$ K.N.)



4/

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x	$-\infty$		0		1		$+\infty$
y'	$-$		$\|\|$	$-$	0	$+$	
y	0	$\searrow$	$-\infty$ $\|\|$ $+\infty$	$\searrow$	e	$\nearrow$	∞

4. min.

4/